

Scientific works in detail

July 30, 2026

Nicolas BOUSQUET

Research webpage: <https://bousquet-n.github.io/research/>

Scientific specializations

Mathematics of risk analysis, Bayesian modeling, surrogate modeling and alignment, artificial intelligence, uncertainty quantification, and decision-making, with major applications in industry, health and the environment.

Grants, honours & awards

2023-present: Expert, Collège France Numérique 2030

2021-present: Expert, Horizon Europe

2019 AMIES PEPS Grant, Lymphomath Project

2016 ENBIS Young European Statistician Award

2014 Best conference paper, Lambda-Mu Conference in Risk and Reliability

2014 IDEX Grant, Metadeb Project, Institut de Mathématique de Toulouse

R&D works / organization

- I authored or co-authored >55 research articles, book chapters and vulgarization papers + 3 patents. I supervised and co-authored 3 scientific textbooks. I have been a long-term referee and associate editor for top-tier peer-reviewed scientific journals. Finally, I organized learning expeditions (LE) and research schools, and was the scientific leader of a special 2019 LE about AI under the authority of the French Ministry of Foreign Affairs and International Development, for the benefit of the French government and the main French industrial companies and laboratories
- I co-supervised several PhD CIFRE students and post-doc researchers, with École Polytechnique, Sorbonne Université, IMT Toulouse, ENSTA, INRAE (*see further for details*)
- I founded the Young Statistician Group and was the head of the Environment Group of the French Statistical Society (SFdS). I organized numerous public conference and workshops for RT-UQ (MASCOT-NUM), SFdS, EDF, Quantmetry, ENBIS, Forum Emploi Math and the UQSay international seminar. I led the organization of the EDF-INRIA-CEA 2025 Summer School on Numerical Analysis ([website](#))
- I have been a recruitment commission member / team referee for INRIA, IFREMER, Fullbright Commission, ANRT, FP7/H2020/Horizon Europe, ANSES, Deutsches Zentrum für Luft und Raumfahrt.
- I was in the scientific committee of French Statistical Days in 2018 & 2022, as for the International Environmetrics Society (TIES ; 2021-2022). I am a long-term member of the Scientific Advisory Board of the *Laboratoire National de Métrologie et d'Essais* (LNE) , from 2020.
- I have been an International Expert for the Food and Agriculture Organization (FAO) and the European Commission (Horizon Europe, FP7 projects) for projects related to biodiversity, halieutic sciences and ecosystemic sciences.

Software development

1. From 2024: supervising the development of Python toolkits for conservative structural risk assessment
2. 2022-2024: supervision of the development of multimodal (RAG-LLM based) MACADAM AI tools for industrial alert systems.
3. 2022-2024: supervision of the pipeline development for guiding Explainable AI in production (EDF Group-).
4. 2017-2020: supervision of the development of the Machine Learning platform at Quantmetry with clients (open source Python codes including `skope-rules` (scikit-contrib) and `melusine`).
5. 2013-2016: co-supervision of the development of the R package `mistral` for accelerated Monte Carlo methods in structural reliability frameworks (EDF).
6. 2010-2016: co-supervision of the development of the MATLAB software `WOLF3` for steel toughness quantification (joint work with CEA).
7. 2009-2011: development of the the R/C/Excel software `SIMCAB` (*Seal Impact on Cod ABundance*) for Fisheries & Oceans Canada (DFO).
8. 2007-2010: development of the software tool `QuANFRE` (XML/R/C) for EDF: estimation of life-time models from censored data in both frequentist and Bayesian frameworks.

Published research works

1. Borel, M., E. Buchoud, J. Mars, C. Ioana, A. Simon, and N. Bousquet (2026). *Global and local feature selection methods for analyzing measurements of dam*. In: *Actes de la Conférence sur l'Apprentissage Automatique (CAP'06)*.
2. Ferrere, B., N. Bousquet, F. Gamboa, J.-M. Loubes, and J. Muré (2026). *Exact Functional ANOVA Decomposition for Categorical Inputs Models*. In: *Proceedings of the International Conference on Machine Learning (ICML 2026)*, Seoul, Korea. Spotlight paper.
3. Kazmierczak, R., S. Azzolin, E. Berthier, A. Hedström, P. Delhomme, N. Bousquet, G. Frehse, M. Mancini, B. Caramiaux, A. Passerini, and G. Franchi (2026). *Benchmarking XAI Explanations with Human-Aligned Evaluations*. In: *Proceedings of the 40th Annual Conference on Artificial Intelligence (AAAI'26)*.
4. Ketema, B., N. Bousquet, F. Costantino, F. Gamboa, B. Iooss, and R. Sueur (2025a). *Geodesic Non-completeness of the Truncated Normal Family*. In: *Geometric Science of Information: 7th International Conference, GSI 2025*. Vol. 2. Berlin, Heidelberg: Springer-Verlag, pp. 32–40.
5. — (2025b). *Fisher-Rao distance between truncated distributions and robustness analysis in uncertainty quantification*. In: *Journal of Statistical Computation and Simulation* 95.17, pp. 3633–3659.
6. Bousquet, N., M. Blazère, and T. Cerbelaud (2025). *Covariance constraints for stochastic inverse problems of computer models*. In: *Electronic Journal of Statistics* 19.1, pp. 1809–1854.
7. Il Idrissi, M., N. Bousquet, F. Gamboa, B. Iooss, and J.-M. Loubes (2025). *Hoeffding decomposition of functions of random dependent variables*. In: *Journal of Multivariate Analysis* 208, p. 105444.
8. — (2024). *Quantile-constrained Wasserstein projections for robust interpretability of numerical and machine learning models*. In: *Electronic Journal of Statistics* 18.2, pp. 2721–2770.
9. Bousquet, N. (2024). *Specifying prior distributions in reliability applications: Towards new formal rules for informative prior elicitation?* In: *Applied Stochastic Models in Business and Industry* 40.1, pp. 92–102.
10. Ouzoulis, F., N. Bousquet, M. Genu, A. Gilles, J. Spitz, and M. Authier (2024). *Development of a new control rule for managing anthropogenic removals of protected, endangered or threatened species in marine ecosystems*. In: *PeerJ*. 12:e16688.
11. Ketema, B., N. Bousquet, F. Costantino, F. Gamboa, B. Iooss, and R. Sueur (2024). *Non-asymptotic confidence intervals for importance sampling estimators of quantiles*. In: *Act. J. Statist.*

12. Il Idrissi, M., N. Bousquet, F. Gamboa, B. Iooss, and J.-M. Loubes (2023). *On the coalitional decomposition of parameters of interest*. In: *Comptes-Rendus de l'Académie des Sciences* 361, pp. 1653–1662.
13. Simoulin, A., N. Thiebaut, K. Neuberger, I. Ibnouhsein, N. Brunel, R. Viné, N. Bousquet, J. Latapy, N. Reix, Molière, L. Lodi, and C. Mathelin (2023). *From free-text electronic health records to structured cohorts: Onconum, an innovative methodology for real-world data mining in breast cancer*. In: *Computer Methods and Programs in Biomedicine* 240, p. 107693.
14. Blanchard, J.-B., R. Chocat, G. Damblin, M. Baudin, N. Bousquet, V. Chabridon, B. Iooss, M. Keller, J. Pelamatti, and R. Sueur (2023). *Fiche pédagogique sur le traitement des incertitudes dans les codes de calcul*. Tech. rep. Tripartite Institute EDF-CEA-FRAMATOME.
15. Il Idrissi, M., N. Bousquet, F. Gamboa, B. Iooss, and J.-M. Loubes (2022). *Projection de mesures de probabilité sous contraintes de quantile par distance de Wasserstein et approximation monotone polynomiale*. In: *Act. J. Statist.*
16. Béthencourt, L., W. Dabachine, V. Dejoui, Z. Lalmiche, K. Neuberger, I. Ibnouhsein, S. Chéreau, C. Mathelin, N. Savy, P. Saint Pierre, and N. Bousquet (2021). *Guiding Measurement Protocols of Connected Medical Devices Using Digital Twins: A Statistical Methodology Applied to Detecting and Monitoring Lymphedema*. In: *IEEE Access* 9, pp. 39444–39465.
17. Bousquet, N. (2021). *Bayesian Extreme Value Theory*. In: *Statistical Extreme Value Theory with Applications to Natural Hazard Engineering*. Springer Nature: Heidelberg.
18. Roussel, J., S. Ancelet, S. Djazoubi, M.-O. Bernier, D. Ricard, and N. Bousquet (2021). *Inférence bayésienne de l'évolution de l'atropie cérébrale et de plages de leucopathie à partir de séquences IRM 3D non homogénéisées*. In: *Act. J. Statist.*
19. R. Adon, R., F. Arthur, G. Baquias, G. Hochard, A. Kaid Gherbi, A. Nègre, A. Simoulin, F. Talaoui-Mockli, and N. Bousquet (2020). *Deep Learning : des usages contrastés dans le monde socio-économique. Une contextualisation de l'ouvrage de Goodfellow, Bengio et Courville*. In: *Statistique et Société* 8, pp. 55–108.
20. Benoumechiar, N., B. Michel, P. Saint-Pierre, and N. Bousquet (2020). *Detecting and modeling critical dependence structures between random inputs of computer models*. In: *Dependence Modeling* 8, pp. 263–297.
21. Bousquet, N., T. Klein, and V. Moutoussamy (2018). *Approximation of limit state surfaces in monotonic Monte Carlo settings, with applications to classification*. In: *SIAM Journal on Uncertainty Quantification* 6, pp. 1–33.
22. Bousquet, N. (2018a). *Modélisation bayésienne des extrêmes*. In: *Événements naturels extrêmes. Théorie statistique et mitigation du risque*. Lavoisier.
23. Parey, S., T. Hoang, and N. Bousquet (2018). *Modèles stochastiques et physiques pour la simulation*. In: *Événements naturels extrêmes. Théorie statistique et mitigation du risque*. Lavoisier.
24. Sonigo, C., S. Jankowski, O. Yoo, O. Trassard, N. Bousquet, M. Grynberg, I. Beau, and N. Binart (2018a). *High-throughput ovarian follicle counting by an innovative deep learning approach*. In: *Scientific Reports* 8.
25. — (2018b). *Comptage des follicules primordiaux par deep learning : l'intelligence artificielle au service de l'étude de la reproduction*. In: *Annales d'Endocrinologie* 79.
26. Bousquet, N. (2018b). *Modeling Extreme Events in Energy Companies*. In: *Encyclopedia of Statistics (Wiley StatRef)*.
27. Pérot, N. and N. Bousquet (2017). *Functional Weibull-based models of steel fracture toughness for structural risk analysis: estimation and selection*. In: *Reliability Engineering and System Safety* 165, pp. 355–367.
28. Fu, S., M. Couplet, and N. Bousquet (2017). *An adaptive kriging method for solving nonlinear inverse statistical problems*. In: *Environmetrics* 4, e2439.
29. Wolfson, L. and N. Bousquet (2016). *Elicitation*. In: *Encyclopedia of Statistics (Wiley StatRef)*.

30. Sueur, R., N. Bousquet, B. Iooss, and J. Bect (2016). *Perturbed-Law based sensitivity analysis indices in structural safety*. In: *Proceedings of SAMO'16*.
31. Lemaître, P., E. Sergienko, A. Arnaud, Bousquet, N., F. Gamboa, and B. Iooss (2015). *Density modification based reliability sensitivity indices*. In: *Journal of Statistical Computation and Simulation* 85, pp. 1200–1223.
32. Bousquet, N., M. Fouladirad, A. Grall, and C. Paroissin (2015). *Bayesian gamma processes for optimizing condition-based maintenance under uncertainty*. In: *Applied Stochastic Models in Business and Industry* 31.3, pp. 360–379.
33. Keller, M., A. Popelin, N. Bousquet, and E. Remy (2015). *Nonparametric estimation of the probability of detection of flaws in an industrial component, from destructive and nondestructive testing data using Approximate Bayesian Computation*. In: *Risk Analysis* 35, pp. 1595–1610.
34. Dortel, E., F. Sardenne, N. Bousquet, E. Rivot, J. Million, G. Le Croizier, and E. Chassot (2015). *An integrated Bayesian modeling approach for the growth of Indian Ocean yellowfin tuna*. In: *Fisheries Research*.
35. Fu, S., G. Celeux, N. Bousquet, and M. Couplet (2015). *Bayesian inference for inverse problems occurring in uncertainty analysis*. In: *International Journal for Uncertainty Quantification* 5, pp. 73–98.
36. Pasanisi, A., C. Roero, E. Remy, and N. Bousquet (2015). *On the practical interest of discrete Inverse Polya and Weibull-1 models in industrial reliability studies*. In: *Quality and Reliability Engineering International* 31, pp. 1161–1175.
37. Bousquet, N. and F. Douard (2014). *Analyse bayésienne d'intensités de défaillance pour les études de gestion d'actif*. In: *Act. Lambda-Mu*. Best paper award.
38. Parent, E., J. Bernier, A. Pasanisi, N. Bousquet, and M. Keller (2014). “Considérations décisionnelles pour la construction d’un ouvrage de protection contre les crues”. In: *Approches Statistiques du Risque*. Technip.
39. Bect, J., N. Bousquet, B. Iooss, S. Liu, A. Mabilie, A.-L. Popelin, T. Rivière, R. Stroh, R. Sueur, and E. Vazquez (2014). *Uncertainty quantification and reduction for the monotonicity properties of expensive-to-evaluate computer models*. In: *Proceedings of UCM 14, Sheffield*.
40. Archambault, B., O. Le Pape, N. Bousquet, and E. Rivot (Jan. 2014). *Density-dependence can be revealed by modelling the variance in the stock-recruitment process: an application to flatfish*. In: *ICES Journal of Marine Science* 71.8, pp. 2127–2140.
41. Bousquet, N., E. Chassot, D. Duplisea, and M. Hammill (2014). *Forecasting the major influences of predation and environment on cod recovery in the northern Gulf of St. Lawrence*. In: *Plos ONE* 9, e82836.
42. Moutoussamy, V., N. Bousquet, B. Iooss, P. Rochet, T. Klein, and F. Gamboa (2013). *Comparing conservative estimations of failure probabilities using sequential designs of experiments in monotone frameworks*. In: *11th International Conference on Structural Safety & Reliability (ICOSSAR), June 16-20, 2013, New York*.
43. Dortel, E., F. Massot-Granier, E. Rivot, J. Million, J.-P. Hallier, E. Morize, J.-M. Munaron, N. Bousquet, and E. Chassot (2013). *Accounting for Age Uncertainty in Growth Modeling, the Case Study of Yellowfin Tuna (Thunnus albacares) of the Indian Ocean*. In: *PLOS One* 8, e60886.
44. Fouladirad, M., C. Paroissin, N. Bousquet, and A. Grall (2013). *Bayesian optimization of condition-based maintenance under uncertainty*. In: *Proceedings of MMR'13*.
45. Bousquet, N., E. Dortel, E. Chassot, J. Million, J. Eveson, and J.-P. Hallier (2012). *Preliminary assessments of tuna natural mortality rates from a Bayesian Brownie-Petersen model*. In: *Proceedings of IOTC-2012-WPTT14-41*.
46. Pasanisi, A., S. Fu, and N. Bousquet (2012). *Estimating discrete Markov models from various incomplete data schemes*. In: *Computational Statistics and Data Analysis* 56, pp. 2609–2625.
47. Bousquet, N. (2012). *Accelerated Monte Carlo estimation of exceedance probabilities under monotonicity constraints*. In: *Annales de la Faculté des Sciences de Toulouse* 21, pp. 557–591.

48. Bousquet, N. (2010a). *Elicitation of Weibull priors*. In: *Proceedings of MMR'09*.
49. Bousquet, N., N. Cadigan, T. Duchesne, and L.-P. Rivest (2010). *Detecting and correcting under-reported catches in fish stock assessment: trial of a new method*. In: *Canadian Journal of Fisheries and Aquatic Sciences* 67.8, pp. 1247–1261.
50. Bousquet, N. (2010b). *Eliciting vague but proper maximal entropy priors in Bayesian experiments*. In: *Statistical Papers* 51.3, pp. 613–628.
51. Pasanisi, A., S. Fu, and N. Bousquet (2010). *Estimation de modèles markoviens discrets dans un cadre industriel fiable à données manquantes*. In: *Act. J. Statist.*
52. Pasanisi, A., E. de Rocquigny, E. Parent, and N. Bousquet (2009). *Some useful features of the Bayesian setting while dealing with uncertainties in industrial practice*. In: *Proceedings of the European Safety and Reliability Conference (ESREL)*.
53. Chassot, E., D. Duplisea, M. Hammill, A. Caskenette, N. Bousquet, Y. Lambert, and G. Stenson (2009). *Role of predation by harp seals (Pagophilus groenlandicus) in the collapse and non-recovery of northern Gulf of St. Lawrence cod (Gadus morhua)*. In: *Marine Ecology Progress Series* 379, pp. 279–297.
54. Bousquet, N. (2009). *Advantages and challenges of Bayesian statistical analysis in industrial lifetime*. In: *Lettres Techniques de l'Ingénieur: Risques Industriels* 23, pp. 5–8.
55. Bousquet, N., T. Duchesne, and L.-P. Rivest (2008). *Redefining the maximum sustainable yield for the Schaefer population model including multiplicative environmental noise*. In: *Journal of Theoretical Biology* 254, pp. 65–75.
56. Bousquet, N. (2008). *Diagnostics of prior-data agreement in applied Bayesian analysis*. In: *Journal of Applied Statistics* 35, pp. 1011–1029.
57. Bousquet, N., H. Bertholon, and G. Celeux (2006). *An alternative competing risk model to the Weibull distribution in lifetime data analysis*. In: *Lifetime Data Analysis* 12, pp. 481–504.
58. Bousquet, N. (2005). *Introducing a new way of eliciting prior distributions for Weibull inference in an industrial context*. In: *Communications in Dependability and Quality Management* 8, pp. 23–29.

Doctoral dissertations

1. Bousquet, N. (2024). *Contributions to the statistical quantification of uncertainties affecting the use of numerical models*. Habilitation dissertation. Sorbonne Université’.
2. — (2006). *A methodology of Bayesian analysis for the lifetime prediction of industrial components*. PhD thesis. Université Paris-XI (Paris-Saclay).

Scientific books

1. Bousquet, N. and P. Bernardara (eds) (2021). *Extreme Value Theory with Applications to Natural Hazard Engineering*. Springer Nature: Heidelberg.
2. — (2018). *Événements extrêmes naturels : théorie statistique et mitigation du risque*. Lavoisier: Paris.
3. Goodfellow, I., Y. Bengio, and A. Courville (2018). *L'apprentissage profond. Supervisor of the English-to-French translation* $\Leftrightarrow$ the first scientific textbook translated using an AI tool mixing regex and deep learning techniques. Éditions Florent Massot.

Submitted research works

1. Ferrere, B., N. Bousquet, F. Gamboa, J.-M. Loubes, and J. Muré (2026). *Fourier Analysis on the Boolean Hypercube via Hoeffding Functional Decomposition*. In: *Submitted to TMLR*.
2. Ferrere, B., N. Bousquet, F. Gamboa, and J.-M. Loubes (2026). *Generalized Functional ANOVA in Closed-Form: A Unified View of Additive Explanations*. In: *Submitted to NeurIPS 2026*.
3. Ancelet, S., N. Bousquet, and E. Parent (2026). *Optimal nonlinear Bayesian design for structural reliability estimation under experimental cost constraints*. In: *submission*.

4. Bousquet, N (2026). *Computing conservative probabilities of rare events with surrogates*. In: *submission*. arXiv:2403.17505.
5. Rives, G., O. Lopez, and N. Bousquet (2025). *WTNN: Weibull-Tailored Neural Networks for survival analysis*. In: *Submitted to Lifetime Data Analysis*. arXiv:2512.09163.
6. Medernach, D., J. Richon, N. Bousquet, C. Sakarovitch, M. Suassuna Santos, S. Worbe, M. Heckmann, and N. Brunel (2025a). *Bridging the Gap between Streamflow Forecasting and Water Allocation: A Two-Phase Challenge in Water Scarcity Hackathon*. In: *submission*.
7. — (2025b). *Methodology for Water Scarcity Hackathon: A Two-Phase Framework for Water Allocation under Uncertain Streamflow*. In: *submission*.
8. Ketema, B., N. Bousquet, F. Costantino, F. Gamboa, B. Iooss, and R. Sueur (2024). *On Nature's Subproblem for Quantile Risk Measures and Parametric Ambiguity Sets*. In: *submitted*.

Patents

1. N. Pérot, N. Bousquet, M. Marques (2015). Method for determining the strength distribution and the ductile-brittle transition temperature of a steel product subjected to thermal variations. *European Patent WO-2015-165962*.
2. D. Vautrin, N. Bousquet, E. de Oliveira, K. Zoubert-Ousseni, N. Paul (2016). Managing water-supply pumping for an electric production plant circuit. *European Patent WO-2017-86603*.
3. N. Bousquet, A. Guillaume, G. Franchi (2025). Method for determining data representative of a risk of occurrence of a clogging event in a pumping station. *European Patent Deposit PCT/EP2025/082315*.

Other scientific contributions (Selected EDF research papers with restricted access)

1. N. Bousquet (2009). A statistical approach of steel (16MND5) toughness variation in function of temperature. EDF R&D H-T56-2009-02278-FR.
2. N. Bousquet (2009). Quantifying the input uncertainty for computer codes used in nuclear safety assessments : the case for manufacturing defect sizes. EDF R&D H-T56-2009-02088-FR.
3. N. Bousquet (2010). The Monotonic Reliability Method. Application to hydraulic computer models. EDF R&D H-T56-2011-00261-FR.
4. N. Bousquet (2010). Accounting for the ultrasonic probability detection in the assessment of input uncertainty affecting defect sizes in nuclear safety assessments. EDF R&D H-T56-2011-00348-FR.
5. N. Bousquet (2011). Probabilistic calibration of rating curves in hydraulics. EDF R&D H-T56-2011-02059-FR.
6. N. Bousquet (2011). Mathematical improvements on the Monotonic Reliability Method, with application to hydraulic computer models. EDF R&D H-T56-2011-02057-FR.
7. M. Keller, N. Bousquet (2012). Joint statistical estimation of defect size and density distributions in steel structures, submitted to observational and detection errors. EDF R&D H-T56-2011-02379-FR.
8. N. Bousquet (2012). Defining effective sample sizes for censored data. EDF R&D H-T56-2011-02069-FR.
9. N. Bousquet (2012). A Bayesian hierarchical model for Common Cause Failures in Probabilistic Safety Assessment. EDF R&D H-T56-2012-02491-FR.
10. N. Bousquet, M. Couplet, S. Fu (2012). A Bayesian methodology for the inverse calibration of Strickler-Manning roughness coefficients in hydraulic computer models, with application to Garonne river using TELEMAC-2D code. EDF R&D H-T56-2012-02079-FR.
11. N. Bousquet (2013). A probabilistic methodology for quantifying the risk of partial and total loss of water cooling systems on coastal nuclear plants. EDF R&D H-T55-2013-0561-FR.
12. P. Braghini, B. Iooss, N. Bousquet (2013). Designing algorithms for exploring monotonic computer models prior to the use of accelerated Monte Carlo methods. EDF R&D H-T55-2013-03339-FR.

13. N. Bousquet (2014). A copula-based approach to assess the difficulty and intervention time of nuclear plant operators in deteriorated conditions. EDF R&D H-T55-2014-00594-FR.
14. P. Lemaître, B. Iooss, A. Arnaud, N. Bousquet (2014). Sensitivity analysis in structural reliability studies : a review. EDF R&D H-T57-2014-03082-FR.
15. K. Zoubert, N. Bousquet, D. Vautrin (2014). Statistical machine learning for the development of alert and monitoring systems for nuclear plants face to massive clogging. EDF R&D H-T55-2014-08101-FR.
16. N. Bousquet, T. Papelard (2015). A complete probabilistic methodology for quantifying the risk of partial and total loss of water cooling systems for the French nuclear park. EDF R&D H-T55-2015-02318-FR.
17. N. Bousquet, C. Denis (2017). Elements of justification for the use of probability in the modeling of epistemic uncertainties affecting black-box computer models. EDF R&D 6125-1614-2017-00965-FR.
18. N. Bousquet, V. Le Guen, A. Ribes (2021). Hybrid metamodeling for fields physics: a critical review and perspectives in CFD and electromagnetism. EDF R&D 6125-3119-2021-03069-EN.
19. N. Bousquet, L. Apffel, B. Iooss (2020). Roadmap of the AI SINCLAIR Laboratory. EDF R&D 6125-3114-2020-03850-FR.
20. N. Bousquet (2021). Formalization of the voltage drop detection problem predicted by the FluxBT tool and proposals for the implementation. EDF R&D 6125-3119-2021-02102-FR.
21. N. Bousquet, B. Iooss, V. Chabridon, J. Pelamatti, J. Muré, M. Il Idrissi C. Poinson, P. Bailly (2022). A methodological and operational guide to AI interpretability. EDF R&D 6125-3119-2022-03068-EN.
22. L. Ramiandrisoa, N. Bousquet, M. Baudin, B. Iooss, C. Poinson, P. Bailly (2023). Lexique commun aux domaines de la métrologie, des statistiques et de la modélisation sur le thème des "incertitudes". EDF R&D 6125-3115-2023-01445-FR.
23. B. Iooss, v. Chabridon, N. Bousquet, M. Keller, J. Pelamatti, R. Sueur (2023). Introduction au traitement des incertitudes dans les codes de calcul. Fiche I3P BIAS (EDF, CEA, FRAMATOME).
24. I. Seydi, J. Pelamatti, N. Bousquet (2024). Etude de méthodes de classification interprétables à base de projection par shapelets. Application à l'identification d'hydrogrammes de crue non plausibles.

Selected lectures and keynotes

1. Bousquet, N. (2026). Implementing AI in industry: some lessons, bottlenecks and research avenues. *Water & AI Conference*
2. Bousquet, N. (2025). New results in AI interpretability. *ECML-PKDD 25*, Porto
3. Bousquet, N. (2025). Explainable AI in industry. *Mobilit'AI*
4. Bousquet, N. (2023). *Auditable Bayesian modeling*. *Institut des Actuaire*s
5. Bousquet, N. (2022). *Auditable Bayesian modeling for quality measurements*. *MATHMET Conference*
6. Bousquet, N. (2022). *Risk analysis, uncertainty and robust decision-making: an introduction*. *MASCOT-NUM Workshop on Statistical methods for Safety and Decommissioning*
7. Bousquet, N. (2022). *Copulas in uncertainty quantification*. *MEXICO Network Conference*
8. Bousquet, N. (2022). Decision-helping for stopping machines before rare and fast natural hazards: statistical approaches can beat machine learning rules. *FINS Workshop on "Recent Developments in Dependence Modelling with Applications in Finance and Insurance*, Agistri
9. Bousquet, N. (2021). Powerful deep approaches to follicle counting in fertility studies through medical imaging. *GDR REPRO Days*
10. Bousquet, N. (2020). *Online alert systems againsts rare events affecting devices interacting with the environment*. *TIES Conference*

11. Bousquet, N. (2020). [Connections between sensitivity analysis and well-posed problems in stochastic inversion](#). *Saclay Seminar on Uncertainty Quantification* (UQSay)
12. Bousquet, N. (2019). Statistical calibration of the digital twin of a connected health object. *SIS 2019 Conference*
13. Bousquet, N. (2019). About the geometry of black-box models. *Academia Sinica*, Taipei
14. Bousquet, N. (2018). How treating uncertainty in black-box complex models? *LIP6-IPSL Seminar on Artificial Intelligence and Climate*
15. Bousquet, N. (2017). [Expert elicitation methods](#). Course at *ETICS-17 Summer School*, Porquerolles
16. Bousquet, N. (2017). High-dimensional copulas for solving unbalanced classification problems in industrial risk mitigation. *5th Symposium on Games and Decisions in Reliability and Risk*
17. Bousquet, N. (2016). Convexity and monotonicity in accelerated Monte Carlo methods for structural reliability. *ALT Conference 2016*
18. Bousquet, N. (2016). How geometric properties of black box computer models can help to provide conservative assessments in time-consuming situations. *ENBIS 2016 Conference*
19. Bousquet, N. (2010). Bounding and estimating probabilities of extreme events in natural resource management. *NCIDS Workshop on Statistics using Spatial and Geographical Data*

Ph.D. students

1. Anaïs Durin (Université Grenoble Alpes - Academic co-advisor: G. Fabbri): *The role of technological, economic, and societal levers—and their interactions—in climate-energy foresight scenarios: an analysis using explainable machine learning* (2025-2028).
2. Baptiste Ferrere (Université Toulouse III - Co-advisors: F. Gamboa / JM Loubes): *Analyzing AI tools with decomposition methods* (2024-2027).
3. Gabrielle Rives (Sorbonne Université - Co-advisor: O. Lopez): *Multivariate regression under noisy and strongly censored covariates for mission success forecasting* (2022-2025).
4. Marouane Il Idrissi (Univ. Toulouse III - Co-advisors: JM Loubes / F. Gamboa / B. Iooss): *Unification of sensitivity analysis and post-hoc interpretability methods in machine learning* (2021-2024).
5. Nazih Benoumechiara (Sorbonne Université - Co-advisors: G. Biau/P. Saint-Pierre/B. Michel) : *Sensitivity analysis of reliability indicators involving the quantification of stochastic dependencies* (2015-2018).
6. Vincent Moutoussamy (Univ. Toulouse III - Co-advisor: T. Klein): *Accelerated Monte Carlo computation of failure probabilities and quantiles in monotonic frameworks* (2012-2015).
7. Paul Lemaître (Univ. Bordeaux - Co-advisors: P. Del Moral / B. Iooss): *Global sensitivity analysis of failure probabilities in structural reliability analysis (computer experiments)* (2010-2013).
8. Shuai Fu (University Paris XI - Co-advisors : G. Celeux/M. Couplet) : *Nonlinear stochastic inversion for time-consuming computer codes* (2009-2012)

Msc. students

1. **Msc apprenticeship students in Actuarial Science (ISUP)**: I have been the academic tutor of several students per year, from 2018 (69 in total), mostly on parametric insurance problems. Most actuarial dissertations can be found on <https://www.institutdesactuaires.com/>

- Jean-Baptiste Benielli (2018, Axa GIE): [La dépendance : un état d'incapacité consolidé et permanent ?](#) (*Dependency: a consolidated, permanent state of incapacity?*)
- Qi Zuo (2018, Humanis): Automatisation du calcul des provisions des produits d'assurance "Vie entière" et construction d'une loi de sortie adaptée au portefeuille
- Juste A. Nzondo (2018, Axa): [Mise en place d'un indicateur pour les contrats Prévoyance-Santé en Collectives: Valeur-Client](#) (*Accelerating the transition to disability*)
- Christian Chow (2018, Société Générale): [Utilisation des données télématiques pour l'analyse de la sinistralité automobile](#) (*Using telematics data to analyze automobile claims*)
- Thibault Müller (2019, Crédit Agricole Assurances): [Détermination d'un processus de construction et de suivi de loi Best Estimate et application au produit assurance Emprunteur](#)
- Andrew Bernet-Alinot (2019, CNP Assurances): [Accélération du passage en invalidité](#) (*Accelerating the transition to disability*)
- Chen Xi (Groupama, 2020): [Modélisation et évaluation du risque en incapacité de travail](#) (*Modeling and assessing disability risk*)
- Issam Mezrag (2020, Generali): [Construction d'un zonier en assurance MRH](#) (*Developing a territorial rating model for multi-peril household (homeowners) insurance*)
- Wassil Bentarka (2020, Groupama): [Evaluation des provisions techniques et estimation du risque de réserve en assurance construction](#) (*Valuation of technical reserves and estimation of the related risk in construction insurance*)
- Lou Peltier (2021, BNP Paribas Cardif): [Allocation sectorielle des actifs et optimisation des calculs dans un contexte d'application de scénarios climatiques](#) (*Sectoral asset allocation and optimization of calculations in the context of climate scenarios*)
- Raphaël Fellous (2022, Klesia): [Élaboration d'une table d'entrée et de sortie pour le risque d'incapacité à la conduite](#) (*Drawing up an entry/exit table for the risk of unfitness to drive*)
- Romain Chabert (2022, Deloitte Conseil): [Jugement d'expert, biais cognitifs et volatilité - influences sur le provisionnement en assurance non-vie](#) (*Expert judgement, cognitive bias and volatility - influences on non-life insurance reserving*)
- Lucas Jullia (2022, CA Pacifica): [Provisionnement en assurance IARD à l'aide de la data science](#) (*PC insurance provisioning using data science*)
- Corentin Boyeau (2022, Groupama) : [Évaluation de l'impact du changement climatique sur le risque inondation en France métropolitaine](#) (*Assessment of the impact of climate change on the flood risk in mainland France*)
- Corentin Peron (2023, Guy Carpenter) : [Estimation du coût de la sécheresse en France à l'aide de l'apprentissage automatique des données publiques](#) (*Estimating the cost of drought in France using machine learning of public data*)
- Léo Reichardt (2024, Groupama): [Évaluation de l'incertitude sur paramètres et intégration de données externes dans le calcul du risque de primes](#) (*Evaluation of uncertainty on parameters and integration of external data in the calculation of premium risk*)
- Katia Guettouche (2024, Pacifica): [Construction d'un véhiculier en assurance automobile](#) (*Developing a vehicle rating model for motor (auto) insurance*)

[Additional list of trainee actuaries tutored by year:](#)

2018	Poerani Rousseau (ACPR), Sophie Try (Allianz), Vincent Decamps (Generali), Arnaud Meurin (Allianz); Gabrielle Vila (BNP Paribas Cardif)
2019	Pape Mbow (Axa), Guillaume Boutoille (Axa), Raphaël Regembal (Deloitte), Styven Ponnusamy (Allianz), Valentin Bardoux (CNP Luxembourg), Lokmane Benadjal (SOGECAP), Mahutin Dotonou (Crédit Agricole Assurances)
2020	Maylis Kouek (GALEA), Elisabeth de Vandière (Optimind), Julie Boccara (Optimind), Julien Welterlin (KPMG), Wassil Bentarka (Groupama)
2021	Mickaël Thouet (Crédit Agricole Assurances), Floriane Buffet (Addactis), Guillaume Stasinski (GALEA)
2022	Esteban Conjat (Crédit Agricole Assurances), Julia Massimiani (Malakoff Humanis), Lucas Gilmas (AG2R), Shai Azoulay (MGD)
2023	Laurène Choquer (Willis Tower Watson), Geoffroy Lambolez (Optimind), Maxime Segueni (WTW), Marie Delmouly (PwC Audit), Guilhem Soyez (KLESIA Surveillance Actuarielle), Sirine Bedoui (Covéa), Timothée Bernabé (Cardif)
2024	Céline Krichane (Deloitte), Hanounou Kanoute (Abeille Assurances), Julie Mezoul (SwissLife Prev Santé), Maimouna Ndiaye (Generali Retraite), Malika Ganiev (Generali), Marwane Essaid (ACM), Sophianne Tanguy (Generali), Yassin M. Lopez (BPCE Vie), Younes Kaddouri (Generali Vie)
2025	Alimatou Sonko (CNP), Azou Gales (Swiss Re), Jack Nyawa (Groupama), Fatima Hadad (Actuelia), Skander Kherouf (SeaBird), Paul Barthemely (Actélior), Yacine Larbi (Groupama Gan Vie), Florian Gaudin (CNP), Ayman Oussaid El Amri (Mutuelle des Motards), Diana Railean (Axa)

2. Msc research students:

- (a) Imad El Hassouni (École Polytechnique), jointly with Vincent Chabridon: *Conservative risk assessments using biased machine learning* (2025).
- (b) Baptiste Ferrere (ENSAE): *Estimating generalized Hoeffding-based interpretability indices for machine learning problems* (2024).
- (c) Ibrahim Seydi (Sorbonne Université), jointly with Julien Pelamatti : *Classification of credible flood hydrograms with deep learning* (2024).
- (d) Antoine Lhomme (Mines Nancy), jointly with Charlotte G  ry: *A pipeline for converting LaTeX contributions to a HTML wiki website* (2023).
- (e) Fanny Ouzoulias (La Rochelle Universit  ), jointly with Matthieu Authier: *Stochastic Surplus Production Models for Bycatches Management* (2022).
- (f) Bastien Berg  re (Sorbonne Universit  ) : *Improvements on accelerated Monte Carlo techniques in monotonic frameworks* (2021).
- (g) Lo  c B  thencourt (Ecole Centrale Lyon) : *Calibration of a digital twin of a connected object for medical test assesement* (2019)
- (h) Tangi Sala  n (Ecole Centrale Lille, ENSAE) and Nicolas Peltre (Centrale-Sup  lec): *Links between global sensitivity analysis tools and Shapley-based interpretation methods in machine learning* (2018).
- (i) Samy Djazoubi (ENSAI) : *Unsupervised segmentation of brain tumors from clinical routine data* (2017).
- (j) Fariath Soule (Univ. Lyon 1) : *Estimating noisy extreme value models for natural hazards* (2014).
- (k) Kersane Zoubert-Ousseni (ENSAI) : *Detecting massive plugs in nuclear plants using machine learning* (2014).
- (l) Vincent Moutoussamy (Univ. Paris V): *Adaptive accelerated Monte Carlo methods* (2012).
- (m) Shuai Fu (Univ. Paris VI), jointly with Alberto Pasanasi: *Discrete Markovian models and adaptive MCMC* (2009).
- (n) Amine Bennabi (CNAM) : *Bayesian prior elicitation in structural reliability problems* (2008).

Supervised post-doctoral researchers

- 1. Stefan Franzen (Sorbonne Universit  ) : *Bayesian consistency for regression models with missing covariates* (2025)
- 2. Qiao Chen (  cole Polytechnique - Co-advisor : J. Garnier) : *Objectivation in Bayesian modeling* (2025-2027)
- 3. Antoine Guillaume (ENSTA - Co-advisor : G. Franchi) : *Alert systems trained from noisy and unbalanced multivariate series* (2022-2024)
- 4. M  lanie Blaz  re (Univ. Toulouse III - Co-advisor: J.-M. Loubes) : *Well-posed stochastic inversion problems* (2015-2016).
- 5. Paul Rochet (Univ. Toulouse III - Co-advisor : T. Klein) : *Bounding quantiles in monotonic Monte Carlo settings* (2012-2013).
- 6. Sophie Ancelet (ENGREF - Co-advisor : E. Parent) : *Bayesian optimized designs of physical experiments* (2011-2012).
- 7. Merlin Keller (ENGREF - Co-advisor : E. Parent) : *Bayesian treatment of uncertainties in computer codes* (2010-2011).

Organizational and advising duties

- 1. Scientific Advisory Board, Statistics and Uncertainty Department, *Laboratoire National de M  t  rologie et d'Essais* (LNE) , from 2020.
- 2. The International Environmetrics Society (TIES), 2021-2022
- 3. French Statistical Days (JDS), 2018 & 2022.

Service to the profession

Reviewing

1. PhD defense

• Examiner

- (a) Max Halford (Univ. Toulouse III). *Machine Learning Applied to Large Scale Database Optimisation*, September, 2020
- (b) Marion Belloni (IRSN ; Paris-Saclay University). *Hierarchical Bayesian approach for the estimation of radio-induced risks*, March 2021.
- (c) Sosthène Akia (IRD - Université Montpellier II). *Tuna species interactions in Atlantic-African coasts*, September, 2021.
- (d) Jérémie Donà (Sorbonne Université). *Statistical Learning of Physical Dynamics*, May, 2022.
- (e) Thomas Cerbelaud (Mines Paris Tech). *Automatic building energy model calibration for building performance monitoring during operation*, November, 2023.
- (f) Salim Amoukou (Université Paris Saclay). *Trustworthy Machine Learning: Explainability and Distribution-Free Uncertainty Quantification*, December, 2023.
- (g) Madina Traoré (École Centrale Nantes). *Machine learning methods to improve nuclear power plants manoeuvrability*, April, 2024.
- (h) Théo Defontaine (CERFACS). *Machine learning for flood forecasting : extension of the time frame and little heterogeneous data*, May, 2024.
- (i) Bérénice-Alexia Jocteur (Université Lyon I). *Random forest methods for estimating CATE*, July, 2024.
- (j) Bilel Bensaid (CEA/ University of Bordeaux). *Neural network optimization from a PDE perspective*, September, 2024.
- (k) Paul Liautaud (École Polytechnique). *Study of correlation measurements modeling and machine learning, Bayesian estimation of nuclear parameters*, October, 2024
- (l) Julien Demange-Chryst (Université Toulouse / ONERA). *New methods, based on preferential sampling, for estimating sensitivity indices and probabilities of failure*, November, 2024

• Reviewer

- (a) Clara Carlier (CREST/ENSAE + Renault Group). *Model calibration for digital twins of autonomous vehicle*. October 2024.
- (b) Antoine Van Biesbroeck (CMAP / CEA). *Bayesian approaches for nuclear safety studies*. September 2025.
- (c) Jean-Victor Autran (ENIB / ArianeGroup). *DADDY: Detection by Anticipation of malfunctions in embedded systems*. November 2025.
- (d) Olivier Laurent (ENSTA / Univ. Paris Saclay). *Deep Ensembles for Uncertainty Quantification*. October 2026.
- (e) Arthur Roblin (ASNR / Mines Paris Tech). *for detecting radioactive contamination in alpha spectra*. November 2026.

2. Scientific journals and conference proceeding:

Journal of Machine Learning Research, NeurIPS, ICLR, Computational Statistics and Data Analysis (CSDA), International Conference on Geometric Science of Information, Electronic Journal of Statistics (EJS), Computational Statistics (CS), Journal of Statistical Computation and Simulation (JSCS), ESAIM, Brazilian Journal of Statistics, Bayesian Analysis (BA), IEEE Transactions on Reliability, TEST, Entropy, Stat (*International Statistical Institute*), Ecological Modelling, Statistical Modelling, ICES Journal of Marine Science, Journal of Theoretical Biology, Statistics in Medicine, ASCE-ASME Journal of Risk and Uncertainty in Engineering, Systems, Part A: Civil Engineering, ASME Journal of Mechanical Design, Applied Stochastic Models in Business and Industry, ENBIS.

3. National and international research grants / projects:

- (a) Autumn 2025: Member of the Scientific and Editorial Committee for the 2025 AI For Industry Forum: <https://www.aiforindustry.artefact.com/>

- (b) Spring 2025: Member of the Scientific Committee of the 2nd CapGemini Invent Water Scarcity hackathon: <https://www.codabench.org/competitions/4335/>
- (c) Spring 2024: Member of the Scientific Committee of the 1st CapGemini Invent hydrological hackathon: <https://www.codabench.org/competitions/1567/>
- (d) From 2023: France 2030 projects
- (e) INRIA Team DataShape (*topological data analysis*) (2023-2024)
- (f) INRIA Team Statify (*statistical modeling of systems involving data with a complex structure*) (2022-2023)
- (g) The French-American FullBright Commission (2019-2020)
- (h) The French National Association for Research and Technology (2018-2020)
- (i) Various FP7 and H2020 projects (European Commission ; 2013-2019).
- (j) HORIZON-Europe Evaluator (European Climate, Infrastructure and Environment Executive Agency, 2024)
- (k) French Agency for Food, Environmental and Occupational Health and Safety (ANSES ; 2019).
- (l) Deutsches Zentrum für Luft- und Raumfahrt e. V. (DLR) German Aerospace Center Project Management Agency (2016).

4. Recruitment jury member:

- (a) 2018: INRIA Saclay Recruitment Commission (Full-time junior researchers).
- (b) 2014 : IFREMER Recruitment Commission.

5. Scientific tour:

- (a) Under the authority of the French Ministry of Foreign Affairs and International Development and other institutions, I co-organized and led a mission on artificial intelligence in Russia, in November 2019, including visits of public and private AI laboratories in Moscow, Saint-Petersburg and Novosibirsk, with a team of 8 French academic and industrial specialists (ENIB-CNRS, 3IA ANITI, INRIA, Airbus, Naver Labs, Renault Group, Quantmetry). The aim was to determine the level of technological readiness, training and maturity of the ecosystem in order to favor future French-Russian cooperation in the field of AI. A confidential report was produced for the French government and the main French industrial companies and laboratories.

Organizing

1. 2025 : Lead organizer, CEA-EDF-INRIA Summer School in Numerical Analysis: [Solving partial differential equations in fields physics faster with physics-informed machine learning](#)
2. From 2023 : Director, GIS LARTISSTE (<https://uq-at-paris-saclay.github.io>)
 - Lead organizer, GIS Annual Workshops, 2023-2025
3. April, 2023 : Chairman and co-organizer, (*MASCOT-NUM Conference*), Le Croisic.
4. From September 2021 : Co-organizer of the UQSay (*Uncertainty Quantification in Saclay*) Seminar, with CentraleSupélec and ENS Paris-Saclay.
5. 2020-2022: Co-organizer of the annual EDF Lab Data Science and AI Seminar.
6. March 2020: Co-organizer of the Colloquium on Understanding, predicting and mastering environmental risks using statistics of the French Statistical Society (SFdS): https://www.sfds.asso.fr/fr/environnement_et_statistique/658-journee_risque/
7. 2019-2022: Head of the Environment Group at the French Statistical Society (SFdS)
8. June 2019 Organizer of the first open access Quantmetry R&D Half-day on the Future of Artificial Intelligence.
9. September 2018: Organizer of the Special Session on Machine Learning for automatic diagnosis and decision-making at the ENBIS-18 Conference (Nancy).

10. May 2018 : Co-organizer of the SFdS-DMA Day on Interpretability and Fairness of Machine Learning and Artificial Intelligence : https://www.sfds.asso.fr/fr/data_mining_et_apprentissage/484-actualites/
11. June 2018: Co-organizer with EDF RD and the Young Statisticians Group of the first Data Challenge at the Annual SFdS Congress (Saclay)
12. May 2018: Co-organizer of the Seminar on Transparency and Intelligibility of Artificial Intelligence, on behalf of the French Statistical Society (Institut Henri Poincaré)
13. 2016 - 2018: Co-Head of Working Group "Risk and Uncertainty" within the GdR MASCOT-NUM
14. Jun 2016: Special Workshop on the Calibration of Computer Models for Industry and Environment (September, Paris). Organizer on behalf of the SFdS and the GdR MASCOT-NUM
15. Nov 2015: Organizer of the GDR Mascot-Num Seminar on Complex Model Exploration, Institut Henri Poincaré, Paris: <http://www.gdr-mascotnum.fr/nov15.html>
16. 2012-2013: Co-organizer of the First and Second French career fairs specialized in Mathematics: <http://smai.emath.fr/forum-emploi>
17. 2013-2019: Elected Member and Treasurer of the SFdS Group "Environment".
18. 2012-2014: Member of the Group "Industry and Environment" within the International Society for Bayesian Analysis (ISBA), session co-organizer at the annual congress
19. 2011-2012: Co-founder and Deputy Head of the SFdS Young Statisticians Group
20. 2011: Organizer of the special workshop on the Use of Expert Judgment in Risk Analysis (SFdS - LNE), Paris.

Teaching

1. From 2017: MSc Course: Bayesian Modeling and Computational Statistics (with applications in actuarial science). ISUP. Sorbonne Université.
2. 2019: Invited MsC Course: "Quantifying uncertainties and managing inference problems". Bordeaux University.
3. 2017: Bayesian modelling for the treatment of uncertainties. Professional course at ETICS Research Summer School, Porquerolles.
4. 2014-2015: Junior Graduate Bachelor - "Stochastic uncertainties in engineering". Baranquilla University, Colombia. PEERS-AIRD program.
5. 2012: MSc Course - Special Course on Bayesian Decisional Statistics in Uncertainty Frameworks - University City of Caracas, Venezuela.
6. 2013-2016: MSc Course - Computational Statistics, ENSAI.
7. 2010: Professional course "Accounting for Uncertainties", coorganized with the French Statistical Society (SFdS) (Multivariate statistics, probabilistic inversion, accelerated Monte Carlo methods)
8. 2010-2012: Junior Graduate Bachelor - Bayesian Statistics, University Rennes II.
9. 2008-2011: Level-2 Exploratory Statistics, University Paris Dauphine.
10. 2003-2006: Level-1 Algebra and Numerical Analysis, University Paris XI (Saclay).

Scientific outreach

1. Chabridon, V., Ajenjo, A., Baudin, M., Bousquet, N., Fekhari, E., Iooss, B., Keller, M., Muré, J., Pelamatti, J., Remy, E., Sueur, R. (2023). [UQ and ML in Industry: Current Practices and Challenges in Industrial Applications for Low-Carbon Electricity Production](#). AISSAI Workshop on Artificial Intelligence and the Uncertainty Challenge in Fundamental Physics, November 27th, SCAI, Paris.
2. Bousquet, N., Guggiola, A. (2019). Preface to the White Paper "AI, explain yourself!", Quantmetry: <https://www.quantmetry.com/quant-lab/>

3. Bousquet, N. (2018). Poissons, équations. In: *Manifeste pour l'océan*, with F. Agostini, F. Gabart, N. Hulot, P.-Y. Cousteau *et al.* Éditions Autrement.
4. Bousquet, N. (2018). What is practical artificial intelligence? Interviews with three leaders in business innovation. Blog www.quantmetry.com/blog.
5. Bousquet, N. (2014). For a maximized but sustainable fishing. In: *Brève de maths: Mathématiques pour la planète Terre*. Nouveau Monde Éditions.
6. Bousquet, N. (2014). How sizing the dike? In: *Brève de maths: Mathématiques pour la planète Terre*. Nouveau Monde Éditions.